

Donation Tree 3D Model Brief

Create one donation tree model exported as a single GLB file for Three.js, React, and Bolt.

File

- File name: donation-tree.glb
- Website location: public/models/donation-tree.glb

Scene Setup

- Units: meters
- Tree height: about 5 units tall
- Tree base centered at world origin
- Ground/root base at $Y = 0$
- Tree grows upward along positive Y
- Forward direction: -Z
- Apply all transforms before export
- Model origin should be at the center of the tree base on the ground.
- Model bounds should fit roughly X -2.5 to 2.5, Y 0 to 5.5, Z -2.5 to 2.5.

Required Object Names

- TREE_TRUNK
- TREE_ROOTS
- TREE_CANOPY
- PROGRESS_VINE_01
- PROGRESS_VINE_02
- PROGRESS_VINE_03
- Additional donation vines are welcome, but must continue the pattern PROGRESS_VINE_04, PROGRESS_VINE_05, etc.
- Every vine that should change color must start with PROGRESS_VINE_.

Vine Behavior

- The vines should physically wrap upward around the tree from the ground toward the canopy.
- \$0 of \$1000 = vines are black
- \$100 of \$1000 = bottom 10% of vines becomes colored
- \$500 of \$1000 = bottom 50% becomes colored
- \$1000 of \$1000 = all vines become colored
- The vine geometry needs to rise cleanly along the Y axis. The website code reveals color from low Y to high Y.
- Do not create baked animations for donation progress. The website code handles the color reveal.

Materials

- MAT_Trunk_Bark
- MAT_Roots_Dark
- MAT_Canopy_DarkLeaves
- MAT_Vine_Black
- The vines can use MAT_Vine_Black in Blender. The website will override progress vine material in code.
- Recommended colors: trunk dark brown, roots near black brown, canopy very dark green, vines black or near black.

Geometry Guidelines

- Total triangles: ideally under 80k, max around 120k
- No heavy particle systems unless converted to mesh
- No procedural nodes that require Blender at runtime
- Apply modifiers before export, unless they export reliably to GLB
- Use separate mesh objects for trunk, canopy, roots, and progress vines
- Use actual tube/mesh geometry for vines, not just curves, unless curves are converted to mesh before export.
- Each vine should be one continuous upward path if possible.
- Avoid vine parts that go downward for long stretches, because the website reveal is height-based.

Export Settings

- Format: glTF Binary (.glb)
- Include: Selected Objects
- Transform: default glTF export is fine
- Apply Modifiers: Yes
- Materials: Export
- Animation: Off, unless there are idle ambient animations separate from donation progress
- Compression: Draco optional, but not required

Naming Summary

- The website code will look for objects named PROGRESS_VINE_01, PROGRESS_VINE_02, PROGRESS_VINE_03, or any object whose name starts with PROGRESS_VINE_.
- Those are the only meshes that should receive the donation-progress color reveal.